

Sensitivity analyses – HPT (Czech data)

Robustness checks for scoring, ideology operationalisation, exclusions, and random-slopes

HPT and Extremism project

2026-10-02

Contents

1 Purpose and scope	1
1.1 Setup	2
1.2 Data	2
2 1. Scoring variants for HPT (with POP reversed)	3
3 2. Ideology operationalisations	4
4 3. Exclusions: knowledge outliers & extreme SDR	5
5 4. Mixed models with clustering & random slopes	5
6 5. Sanity plots	9
7 6. Read-outs for prose	10
8 Reproducibility appendix	10

1 Purpose and scope

This file documents **exploratory robustness checks** of our main results. We vary how HPT is scored, how ideology is operationalised, which observations are included, and whether class-level **random slopes** are needed. The goal is to see if substantive conclusions survive reasonable perturbations—**not** to hunt for significance.

HPT scoring follows the Hartmann-Hasselhorn / Huijgen instrument logic; note earlier reports that ROA items can behave inconsistently across samples, motivating ROA-free alternatives here. We also leverage the FR-LF dimensions RD and NS for ideology variants. All results explicitly use **reversed POP items** so that higher scores mean **more contextualised/agent-aware** reasoning.

1.1 Setup

```
# Core packages
library(tidyverse)
library(lme4)
library(lmerTest)
library(broom)
library(broom.mixed)
library(performance)
library(glue)
library(gt)
source("scoring_helpers.R")

# Nice printing
theme_set(theme_bw())
```

1.2 Data

```
# Load the dataset created in 00_data-preparation
normalised_responses <- readRDS("student_responses.RDS")
stopifnot(exists("normalised_responses"))
dat_raw <- normalised_responses

# Cluster identifiers
dat_raw <- dat_raw %>%
  mutate(
    school_id = as.factor(school_id),
    class_label = as.factor(class_label),
    class_id = interaction(school_id, class_label, drop = TRUE)
  )

# Reverse POP (1-4) so higher = more contextualised
POP_rev_items <- paste0("POP", 1:3)
dat_raw <- dat_raw %>%
  mutate(across(all_of(POP_rev_items), ~ 5 - as.numeric(.), .names = "{.col}_rev")) %>%
  mutate(
    HPT_POP_rev = scale_mean(., paste0(POP_rev_items, "_rev"), min_answered = 2),
    HPT_CONT = scale_mean(., paste0("CONT", 1:3), min_answered = 2),
    HPT_ROA = scale_mean(., paste0("ROA", 1:3), min_answered = 2),
    # Canonical composites
    HPT_CTX6 = rowMeans(cbind(HPT_POP_rev, HPT_CONT), na.rm = FALSE),
    HPT_TOT9 = rowMeans(cbind(HPT_POP_rev, HPT_CONT, HPT_ROA),
```

```

    na.rm = FALSE)
)

```

Variable dictionary. KN, POP/ROA/CONT, RD/NS, KSA facets, SDR as per codebook.

2 1. Scoring variants for HPT (with POP reversed)

```

# IMPORTANT: use reversed POP columns in all totals

dat <- dat_raw %>%
  mutate(
    HPT_total_9 = HPT_TOT9,
    HPT_total_8 = scale_mean(., c(paste0("POP", 1:3, "_rev"),
                                   paste0("ROA", 2:3),
                                   paste0("CONT", 1:3)), min_answered = 5),
    HPT_total_6 = HPT_CTX6
  )

# Means & SDs so the reader sees scale location and spread
hpt_desc <- dat %>%
  summarise(
    `9-item (POP_rev + ROA + CONT)` := mean(HPT_total_9, na.rm=TRUE),
    `8-item (drop ROA1)`           := mean(HPT_total_8, na.rm=TRUE),
    `6-item (no ROA)`              := mean(HPT_total_6, na.rm=TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(everything(), names_to = "Score", values_to = "Mean")

hpt_sd <- dat %>%
  summarise(
    `9-item (POP_rev + ROA + CONT)` := sd(HPT_total_9, na.rm=TRUE),
    `8-item (drop ROA1)`           := sd(HPT_total_8, na.rm=TRUE),
    `6-item (no ROA)`              := sd(HPT_total_6, na.rm=TRUE)
  ) %>%
  pivot_longer(everything(), names_to = "Score", values_to = "SD")

hpt_desc_tbl <- left_join(hpt_desc, hpt_sd, by = "Score")

hpt_desc_tbl %>%
  gt() %>%
  fmt_number(columns = c(Mean, SD), decimals = 2) %>%

```

HPT scoring variants (POP reversed): means and SDs

Score	Mean	SD
9-item (POP_rev + ROA + CONT)	2.83	0.49
8-item (drop ROA1)	2.82	0.50
6-item (no ROA)	2.84	0.55

```
tab_header(title = "HPT scoring variants (POP reversed): means and SDs")
```

3 2. Ideology operationalisations

```
dat <- dat %>%
  mutate(
    KN_total = rowSums(across(KN1:KN6), na.rm = TRUE),
    SDR_total = scale_mean(., paste0("SDR", 1:5), min_answered = 4),
    NS_sum = scale_mean(., paste0("NS", 1:3), min_answered = 2),
    RD_sum = scale_mean(., paste0("RD", 1:3), min_answered = 2),
    FRLF_mini = scale_mean(., c(paste0("NS", 1:3),
                                paste0("RD", 1:3)), min_answered = 4),
    KSA_A = scale_mean(., paste0("A", 1:3), min_answered = 2),
    KSA_U = scale_mean(., paste0("U", 1:3), min_answered = 2),
    KSA_K = scale_mean(., paste0("K", 1:3), min_answered = 2),
    KSA_total = scale_mean(., c(paste0("A", 1:3), paste0("U", 1:3),
                                paste0("K", 1:3)), min_answered = 7)
  ) %>%
  mutate(across(c(NS_sum, RD_sum, FRLF_mini, KSA_total, KN_total, SDR_total), scale, .names = "{.col}_z"))

# Show quick reliables for predictors (descriptive only)
ideo_desc <- dat %>% summarise(
  KN_mean = mean(KN_total, na.rm=TRUE), KN_sd = sd(KN_total, na.rm=TRUE),
  SDR_mean = mean(SDR_total, na.rm=TRUE), SDR_sd = sd(SDR_total, na.rm=TRUE),
  NS_mean = mean(NS_sum, na.rm=TRUE), NS_sd = sd(NS_sum, na.rm=TRUE),
  RD_mean = mean(RD_sum, na.rm=TRUE), RD_sd = sd(RD_sum, na.rm=TRUE),
  KSA_mean = mean(KSA_total, na.rm=TRUE), KSA_sd = sd(KSA_total, na.rm=TRUE)
)
ideo_desc %>% gt() %>% tab_header(title = "Predictor summaries (raw scale units)")
```

Predictor summaries (raw scale units)

KN_mean	KN_sd	SDR_mean	SDR_sd	NS_mean	NS_sd	RD_mean	RD_sd	KSA_mean	KSA_sd
3.037543	1.622389	3.015548	0.6252279	2.428322	0.8906106	2.539181	0.8846439	2.857781	0.6248808

4 3. Exclusions: knowledge outliers & extreme SDR

```
# Tukey fence for KN; top 10% for SDR
kn_q <- quantile(dat$KN_total, probs = c(.25, .75), na.rm = TRUE)
kn_iqr <- kn_q[2]-kn_q[1]
kn_low <- kn_q[1] - 1.5*kn_iqr
kn_high<- kn_q[2] + 1.5*kn_iqr

sdr_p90 <- quantile(dat$SDR_total, probs = .90, na.rm = TRUE)

dat <- dat %>%
  mutate(
    excl_KN = KN_total < kn_low | KN_total > kn_high,
    excl_SDR = SDR_total >= sdr_p90,
    keep_all = TRUE,
    keep_excl= !(excl_KN | excl_SDR)
  )

excl_tbl <- tibble(
  Criterion = c("Total N", "Drop KN outliers", "Drop top-10% SDR", "Kept (both rules)"),
  N = c(nrow(dat), sum(dat$excl_KN, na.rm=TRUE), sum(dat$excl_SDR, na.rm=TRUE), sum(dat$keep_excl, na.rm=TRUE))
) %>%
  mutate(Percent = scales::percent(N / first(N)))

excl_tbl %>% gt() %>% tab_header(title = "Exclusion counts and percentages")
```

5 4. Mixed models with clustering & random slopes

```
fit_models <- function(data, hpt_var, ideol_var){
  form0 <- as.formula(glue(
    "{hpt_var} ~ {ideol_var} + KN_total_z + SDR_total_z + (1 | school_id) + (1 | class_id)"
  ))
  form1 <- as.formula(glue(
```

Exclusion counts and percentages

Criterion	N	Percent
Total N	293	100%
Drop KN outliers	0	0%
Drop top-10% SDR	32	11%
Kept (both rules)	251	86%

```

    "{hpt_var} ~ {ideol_var} + KN_total_z + SDR_total_z + (1 | school_id) + (1 + {ideol_var} | class_id)"
  ))
  m0 <- lmer(form0, data = data)
  m1 <- try(lmer(form1, data = data), silent = TRUE)
  if (inherits(m1, "try-error") || isTRUE(isSingular(m1))) m1 <- NULL
  list(m0 = m0, m1 = m1)
}

summarise_model <- function(m){
  fx <- broom.mixed::tidy(m, effects = "fixed", conf.int = TRUE)
  r2 <- performance::r2_nakagawa(m)
  fx %>% mutate(R2_marg = r2$R2_marginal, R2_cond = r2$R2_conditional)
}

hpt_vars <- c("HPT_total_9", "HPT_total_8", "HPT_total_6")
ideol_vars <- c("NS_sum_z", "FRLF_mini_z", "KSA_total_z")

# Full sample
full_grid <- tidyr::expand_grid(hpt = hpt_vars, ideol = ideol_vars) %>%
  mutate(fits = map2(hpt, ideol, ~fit_models(dat %>% filter(keep_all), .x, .y)),
         m0 = map(fits, "m0"),
         m1 = map(fits, "m1"))

# Exclusion sample
excl_grid <- tidyr::expand_grid(hpt = hpt_vars, ideol = ideol_vars) %>%
  mutate(fits = map2(hpt, ideol, ~fit_models(dat %>% filter(keep_excl), .x, .y)),
         m0 = map(fits, "m0"),
         m1 = map(fits, "m1"))

collect_table <- function(grid, label){
  out0 <- grid %>% mutate(t0 = map(m0, summarise_model)) %>% unnest(t0) %>% mutate(model = "RI")
  out1 <- grid %>% filter(!map_lgl(m1, is.null)) %>% mutate(t1 = map(m1, summarise_model)) %>% unnest(t1) %>% mutate(model = "RS")

```

```

  bind_rows(out0, out1) %>% mutate(sample = label)
}

tab_full <- collect_table(full_grid, "Full")

## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.

## Warning: There were 9 warnings in `mutate()`.
## The first warning was:
## i In argument: `t0 = map(m0, summarise_model)`.
## Caused by warning:
## ! Can't compute r-squared. Some variance components equal zero. Your model
##   may suffer from singularity (see `?lme4::isSingular` and
##   `?performance::check_singularity`).
##   Decrease the `tolerance` level to force the calculation of random effect
##   variances, or impose priors on your random effects parameters (using
##   packages like `brms` or `glmmTMB`).
## i Run `dplyr::last_dplyr_warnings()` to see the 8 remaining warnings.

## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.

## Warning: There were 2 warnings in `mutate()`.
## The first warning was:
## i In argument: `t1 = map(m1, summarise_model)`.
## Caused by warning:
## ! Can't compute r-squared. Some variance components equal zero. Your model
##   may suffer from singularity (see `?lme4::isSingular` and
##   `?performance::check_singularity`).
##   Decrease the `tolerance` level to force the calculation of random effect
##   variances, or impose priors on your random effects parameters (using
##   packages like `brms` or `glmmTMB`).
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.

tab_excl <- collect_table(excl_grid, "Exclusions applied")

```

```
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
```

```
## Warning: There were 9 warnings in `mutate()`.
```

```
## The first warning was:
```

```
## i In argument: `t0 = map(m0, summarise_model)`.
```

```
## Caused by warning:
```

```
## ! Can't compute r-squared. Some variance components equal zero. Your model
```

```
## may suffer from singularity (see `?lme4::isSingular` and
```

```
## `?performance::check_singularity`).
```

```
## Decrease the `tolerance` level to force the calculation of random effect
```

```
## variances, or impose priors on your random effects parameters (using
```

```
## packages like `brms` or `glmmTMB`).
```

```
## i Run `dplyr::last_dplyr_warnings()` to see the 8 remaining warnings.
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

```
## Warning: There was 1 warning in `mutate()`.
```

```
## i In argument: `t1 = map(m1, summarise_model)`.
```

```
## Caused by warning:
```

```
## ! Can't compute r-squared. Some variance components equal zero. Your model
```

```
## may suffer from singularity (see `?lme4::isSingular` and
```

```
## `?performance::check_singularity`).
```

```
## Decrease the `tolerance` level to force the calculation of random effect
```

```
## variances, or impose priors on your random effects parameters (using
```

```
## packages like `brms` or `glmmTMB`).
```

```
# Keep only ideology terms + intercept
```

```
tab_models <- bind_rows(tab_full, tab_excl) %>%
```

```
  filter(term %in% c("(Intercept)", "NS_sum_z", "FRLF_mini_z", "KSA_total_z")) %>%
```

```
  mutate(
```

```
    ideol = recode(term, NS_sum_z = "NS (z)", FRLF_mini_z = "FR-LF: RD+NS (z)", KSA_total_z = "KSA-3 total (z)", `(Intercept)` = "(Intercept)"),
```

```
    hpt = recode(hpt,
```

```
      HPT_total_9 = "HPT 9-item (POP_rev + ROA + CONT)",
```

```
      HPT_total_8 = "HPT 8-item (drop ROA1)",
```

```
      HPT_total_6 = "HPT 6-item (no ROA)"
```

```
  )
```

```

) %>%
select(sample, hpt, model, ideol, estimate, conf.low, conf.high, p.value, R2_marg, R2_cond) %>%
arrange(sample, hpt, ideol, model)

# Display as a compact table
(tab_models %>%
  mutate(across(c(estimate, conf.low, conf.high, R2_marg, R2_cond), ~round(., 3)),
    p.value = signif(p.value, 3)) %>%
  gt() %>%
  tab_header(title = "Multilevel models: ideology → HPT (POP reversed; controls: KN, SDR; school + class clustering)") %>%
  tab_spanner(label = "Effect ( and 95% CI)", columns = c(estimate, conf.low, conf.high)) %>%
  cols_label(sample="Sample", hpt="HPT score", model="Model", ideol="Predictor",
    estimate=" ", conf.low="CI low", conf.high="CI high", p.value="p",
    R2_marg="R2 (marg.)", R2_cond="R2 (cond.)"))

```

6 5. Sanity plots

```

dat %>%
  ggplot(aes(NS_sum, HPT_total_9)) +
  geom_point(alpha=.25) + geom_smooth(method="lm", se=TRUE) +
  labs(x="NS (sum)", y="HPT total (9-item, POP_rev)",
    title="Bivariate check (unadjusted): NS vs. HPT total (POP reversed)") +
  theme(plot.title.position="plot")

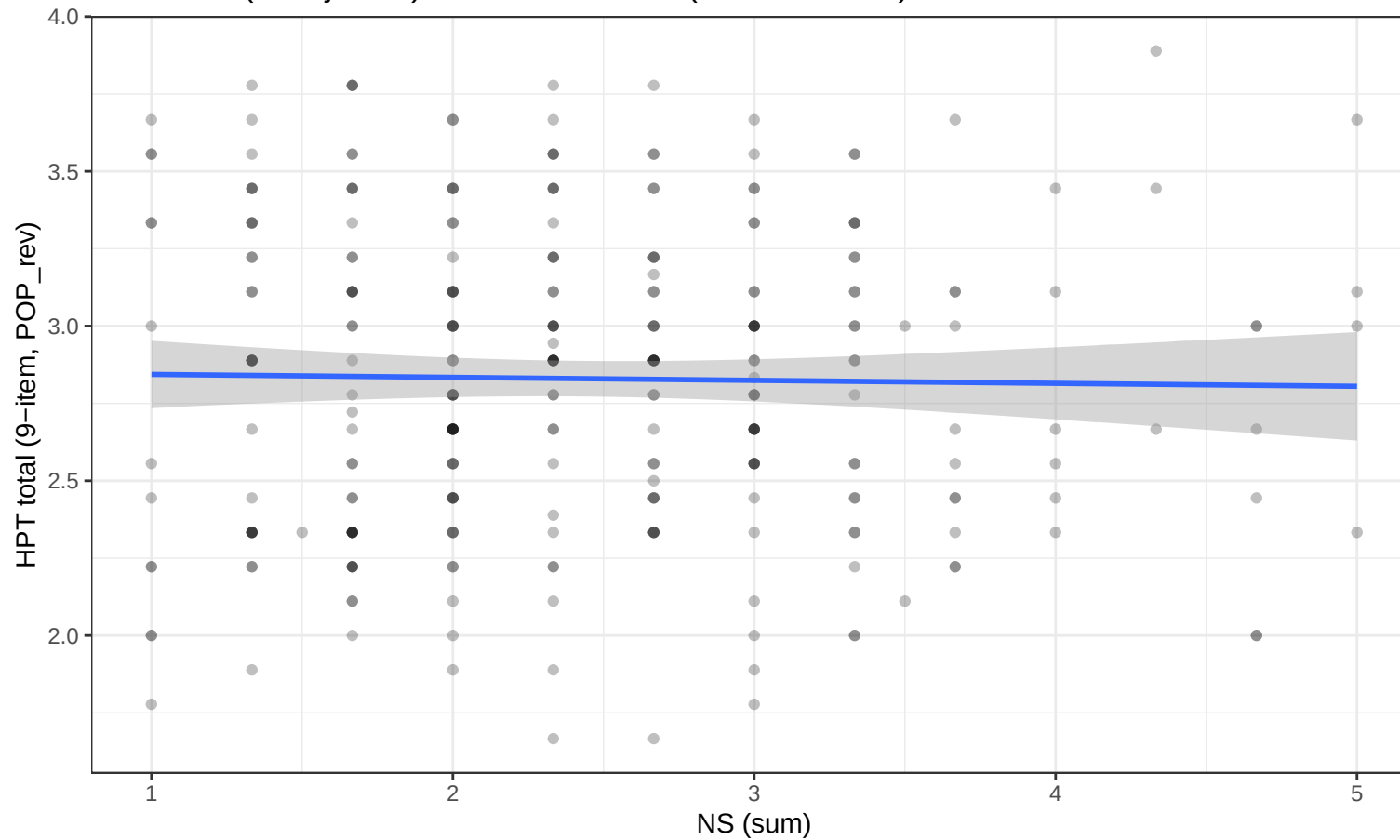
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

Bivariate check (unadjusted): NS vs. HPT total (POP reversed)



7 6. Read-outs for prose

- **Stable conclusions** across HPT 9/8/6 scoring → results **do not depend** on ROA items.
- **NS-only KSA-3** → supports the **ideological contamination** concern.
- Survives **knowledge/SDR exclusions** → less likely driven by misunderstanding or impression management.
- **Random slopes needed** → ideology effects differ **by class** (pedagogical moderation hypothesis).

8 Reproducibility appendix

```
sessionInfo()
```

```

## R version 4.6.1 (2026-06-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0  LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=cs_CZ.UTF-8      LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=cs_CZ.UTF-8  LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=cs_CZ.UTF-8     LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=cs_CZ.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Prague
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
##  [1] gt_1.3.0           glue_1.8.1         performance_0.18.2
##  [4] broom.mixed_0.2.9.7 broom_1.0.13        lmerTest_3.2-1
##  [7] lme4_2.0-6         Matrix_1.7-6       lubridate_1.9.5
## [10] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1
## [13] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2
## [16] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       xfun_0.60          insight_1.5.4
##  [4] lattice_0.23-1     tzdb_0.5.0         numDeriv_2016.8-1.1
##  [7] vctr_0.7.3         tools_4.6.1        Rdpack_2.6.6
## [10] generics_0.1.4     parallel_4.6.1     pkgconfig_2.0.3
## [13] RColorBrewer_1.1-3 S7_0.2.2           lifecycle_1.0.5
## [16] compiler_4.6.1     farver_2.1.2       tinytex_0.61
## [19] codetools_0.2-20   htmltools_0.5.9    yaml_2.3.12
## [22] pillar_1.11.1      furrr_0.4.0        nloptr_2.2.1
## [25] MASS_7.3-66        reformulas_0.4.4    boot_1.3-32
## [28] nlme_3.1-171       parallelly_1.48.0  tidyselect_1.2.1
## [31] digest_0.6.39      stringi_1.8.7      future_1.75.0

```

## [34]	listenv_1.0.0	labeling_0.4.3	splines_4.6.1
## [37]	fastmap_1.2.0	grid_4.6.1	cli_3.6.6
## [40]	magrittr_2.0.5	withr_3.0.3	scales_1.4.0
## [43]	backports_1.5.1	timechange_0.4.0	rmarkdown_2.32
## [46]	globals_0.19.1	otel_0.2.0	hms_1.1.4
## [49]	evaluate_1.0.5	knitr_1.51	rbibutils_2.4.1
## [52]	mgcv_1.9-4	rlang_1.3.0	Rcpp_1.1.2
## [55]	xml2_1.6.0	minqa_1.2.8	R6_2.6.1
## [58]	fs_2.1.0		

Multilevel models: ideology $\rightarrow$ HPT (POP reversed; controls: KN, SDR; school + class clustering)

Sample	HPT score	Model	Predictor	Effect (and 95% CI)			p	R ² (marg.)
				CI low	CI high			
Exclusions applied	HPT 6-item (no ROA)	RI	(Intercept)	2.827	2.715	2.939	2.81e-08	0.124
Exclusions applied	HPT 6-item (no ROA)	RI	(Intercept)	2.826	2.716	2.935	1.80e-08	0.127
Exclusions applied	HPT 6-item (no ROA)	RI	(Intercept)	2.824	2.714	2.933	6.04e-09	0.128
Exclusions applied	HPT 6-item (no ROA)	RI	FR-LF: RD+NS (z)	-0.024	-0.090	0.043	4.85e-01	0.127
Exclusions applied	HPT 6-item (no ROA)	RI	KSA-3 total (z)	-0.033	-0.100	0.033	3.25e-01	0.128
Exclusions applied	HPT 6-item (no ROA)	RI	NS (z)	0.007	-0.061	0.075	8.36e-01	0.124
Exclusions applied	HPT 8-item (drop ROA1)	RI	(Intercept)	2.801	2.692	2.910	3.60e-09	0.147
Exclusions applied	HPT 8-item (drop ROA1)	RI	(Intercept)	2.800	2.692	2.907	2.18e-09	0.150
Exclusions applied	HPT 8-item (drop ROA1)	RI	(Intercept)	2.799	2.690	2.908	1.65e-09	0.150
Exclusions applied	HPT 8-item (drop ROA1)	RS	(Intercept)	2.800	2.690	2.910	2.93e-09	0.146
Exclusions applied	HPT 8-item (drop ROA1)	RI	FR-LF: RD+NS (z)	-0.022	-0.082	0.037	4.62e-01	0.150
Exclusions applied	HPT 8-item (drop ROA1)	RI	KSA-3 total (z)	-0.023	-0.082	0.036	4.49e-01	0.150
Exclusions applied	HPT 8-item (drop ROA1)	RI	NS (z)	-0.004	-0.065	0.056	8.91e-01	0.147
Exclusions applied	HPT 8-item (drop ROA1)	RS	NS (z)	0.000	-0.075	0.074	9.91e-01	0.146
Exclusions applied	HPT 9-item (POP_rev + ROA + CONT)	RI	(Intercept)	2.795	2.677	2.913	1.18e-09	0.148
Exclusions applied	HPT 9-item (POP_rev + ROA + CONT)	RI	(Intercept)	2.794	2.677	2.910	1.02e-09	0.150
Exclusions applied	HPT 9-item (POP_rev + ROA + CONT)	RI	(Intercept)	2.794	2.676	2.911	8.66e-10	0.149
Exclusions applied	HPT 9-item (POP_rev + ROA + CONT)	RI	FR-LF: RD+NS (z)	-0.014	-0.073	0.045	6.40e-01	0.150
Exclusions applied	HPT 9-item (POP_rev + ROA + CONT)	RI	KSA-3 total (z)	-0.011	-0.069	0.048	7.24e-01	0.149
Exclusions applied	HPT 9-item (POP_rev + ROA + CONT)	RI	NS (z)	0.001	-0.059	0.061	9.70e-01	0.148
Full	HPT 6-item (no ROA)	RI	(Intercept)	2.832	2.724	2.940	5.23e-07	0.112
Full	HPT 6-item (no ROA)	RI	(Intercept)	2.833	2.725	2.941	1.04e-06	0.112
Full	HPT 6-item (no ROA)	RI	(Intercept)	2.832	2.727	2.937	3.98e-07	0.115
Full	HPT 6-item (no ROA)	RI	FR-LF: RD+NS (z)	-0.009	-0.071	0.054	7.85e-01	0.112
Full	HPT 6-item (no ROA)	RI	KSA-3 total (z)	-0.029	-0.091	0.033	3.56e-01	0.115
Full	HPT 6-item (no ROA)	RI	NS (z)	0.013	-0.050	0.077	6.76e-01	0.112
Full	HPT 8-item (drop ROA1)	RI	(Intercept)	2.803	2.703	2.902	1.02e-08	0.138
Full	HPT 8-item (drop ROA1)	RI	(Intercept)	2.803	2.704	2.901	9.05e-09	0.139
Full	HPT 8-item (drop ROA1)	RI	(Intercept)	2.802	2.704	2.900	4.73e-09	0.141
Full	HPT 8-item (drop ROA1)	RS	(Intercept)	2.802	2.703	2.900	5.05e-09	0.142
Full	HPT 8-item (drop ROA1)	RI	FR-LF: RD+NS (z)	-0.007	-0.063	0.049	8.03e-01	0.139
Full	HPT 8-item (drop ROA1)	RI	KSA-3 total (z)	-0.023	-0.078	0.033	4.21e-01	0.141
Full	HPT 8-item (drop ROA1)	RS	KSA-3 total (z)	-0.024	-0.089	0.041	4.42e-01	0.142
Full	HPT 8-item (drop ROA1)	RI	NS (z)	0.003	-0.053	0.059	9.12e-01	0.138